

Editorial and Scientific Assessment Report

Molecular Docking Chapter

Overall Assessment

The chapter provides a broad introductory overview of molecular docking and structure-based drug design (SBDD). The conceptual flow—from molecular recognition through historical development, docking methodology, limitations, and future directions—is appropriate for a beginner-oriented textbook. However, several issues should be addressed before publication. These include language quality problems, factual updates, stylistic inconsistencies, and some derivative phrasing typical of review literature. The chapter is scientifically sound overall but requires significant editorial refinement for clarity and accuracy.

Plagiarism Risk Evaluation

No direct verbatim copying was detected in the text. However, multiple passages resemble common formulations found in well-known docking review articles such as Kitchen et al. (2004), Pantsar and Poso (2018), and Sousa et al. (2006). Statements describing the challenges of docking accuracy and the standard three objectives of scoring functions appear to follow conventional phrasing used widely in the literature. While this does not constitute plagiarism, it would be advisable to rewrite some sentences to ensure stronger originality of expression.

Indicators of AI-Assisted Writing

Some sections display characteristics commonly associated with AI-assisted drafting. These include repetitive explanatory structures, uneven conceptual depth between sections, and occasional informal rhetorical statements (e.g., comparisons between human thinking and computers). These stylistic patterns do not invalidate the content but suggest that editorial polishing is necessary to ensure consistency of tone throughout the chapter.

Scientific Accuracy Review

Most of the scientific concepts presented are correct, but several details require revision. The number of structures in the Protein Data Bank is outdated and should be updated to reflect current statistics (over ~230,000 entries as of 2026). Some claims about docking accuracy (e.g., specific success rates attributed to AutoDock Vina) should be presented more cautiously because performance varies depending on datasets and preparation protocols. Statements about entropy treatment in scoring functions should clarify that entropy is often approximated implicitly rather than entirely ignored.

Structural and Pedagogical Review

The chapter structure is logical and suitable for an introductory textbook. However, the introduction is somewhat informal and could benefit from a more academic tone. The historical section is longer than necessary for an opening chapter and may be condensed. Greater pedagogical clarity could be achieved by including conceptual figures, simplified explanatory boxes for key terms, and a few real-world drug discovery examples.

Language and Grammar Issues

Numerous typographical and grammatical errors occur throughout the manuscript. Examples include spelling mistakes, awkward sentence constructions, and inconsistent terminology. The manuscript requires careful copy-editing to correct typographical errors and improve readability. Examples include 'Now a days' (should be 'Nowadays'), 'Depsite effects' ('Despite efforts'), and several other similar errors.

Readability for Broad Audiences

While the chapter is suitable for graduate-level readers, beginners may struggle with some explanations. Readability could be improved by introducing key thermodynamic equations (such as the binding free energy relationship), defining technical terms more carefully, and adding conceptual illustrations of docking workflows, sampling versus scoring, and ligand–receptor interactions.

Strengths of the Chapter

Despite the issues noted above, the chapter has several strong features. It provides a comprehensive overview of molecular docking fundamentals, includes appropriate references to landmark studies, and appropriately emphasizes the limitations of computational predictions. The discussion of emerging areas such as machine learning and cryo-EM also places the field in a modern scientific context.

Editorial Score Summary

Scientific accuracy: 7/10 Originality of phrasing: 7/10 Readability and clarity: 5/10 Language and grammar: 4/10 Pedagogical clarity: 7/10 Overall evaluation: The chapter is scientifically sound but requires moderate editorial revision before publication.